A review of basic spin dynamics

1. Spins in magnetic fields

(a) Consider a single electron spin in a magnetic field $\mathbf{B} = (B_x, B_y, B_z)$. Describe the stationary states (a.k.a energy eigenstates) of the electron spin: what are their energies? What do these states look like in real space? On the Bloch sphere? How can you relate the real space picture to the Bloch sphere picture?

(b) How big are typical values for the energies you calculated in (a) for attainable laboratory magnetic fields? Express these energies in terms of frequency (i.e. energy/h) and temperature (i.e. energy/k_B). For reference, earth's magnetic field is about 0.5 Gauss; a moderate-size air-cooled coil of wire will easily reach 100 G, a strong permanent magnet can create fields of 1000-10,000 G, and a superconducting magnet can exceed 10^5 G. What if the spin was a nuclear spin instead of an electron spin?

(c) Consider a single spin-1/2 in a magnetic field $\mathbf{B} = (B_x, 0, B_z)$. Write down the Hamiltonian for the spin as a 2x2 matrix in the basis of S_z eigenstates. Suppose that at time t = 0 the electron spin starts out "spin up" along the z direction, i.e. in an eigenstate of $\hat{S}_z$ with eigenvalue $\hbar m_s = \hbar/2$. Some time later ($t = \tau$), what is the probability to measure that the spin is still "spin up"? Using the Bloch sphere picture, illustrate how the spin behaves for $B_x >> B_z$, $B_x \sim B_z$, and $B_x << B_z$.

(d) Now consider a spin 1 system. Unlike spin 1/2, spin 1 is not constrained to have degenerate energies in zero magnetic field; for example, the NV center is a spin 1 system for which the Hamiltonian looks like

$$\hat{H} = \Delta S_z^2 - \gamma \mathbf{S} \cdot \mathbf{B}, \tag{1}$$

with $(\Delta/h) = 2.87$ GHz (the "zero field splitting" or ZFS) and $\gamma/h = -2.8$ MHz/Gauss (the "gyromagnetic ratio"). Note that there is a special direction (z) that corresponds to the symmetry axis of the system. Note also that the spin-1 Pauli operators are written as 3x3 matrices. Using your favorite software, diagonalize this Hamiltonian and plot the energy of the eigenstates (in GHz) as a function of magnetic fields from 0 to 1050G for (i) $\mathbf{B}||z$ (ii) $\mathbf{B} \perp z$ and (iii) $\mathbf{B} = B * (0.1, 0, 0.995)$ – i.e. mostly in the z direction.[1] What happens to the energy eigenstates as the magnetic field increases for each situation?

(e) Can you use the Bloch sphere picture for a spin 1? Explain.

2. The rotating wave approximation and Rabi oscillations

This problem considers the interaction between a spin and oscillating magnetic fields, however it looks exactly the same as the interaction between an atomic dipole and electric fields. By mapping atomic states onto spin levels and electric fields onto magnetic fields, you can use an atomic physics textbook to guide you through these problems.

(a) Suppose that two magnetic fields are applied to a spin-1/2 system with gyromagnetic ratio γ: the first magnetic field ($\mathbf{B_0}$) is static in time, while the second magnetic field $\mathbf{B}_{\mathrm{AC}}$ is perpendicular to it and oscillating in time with angular frequency ν, i.e. $\mathbf{B}_{\mathrm{AC}} = \mathbf{B_1} \cos \nu t$. Choosing $z||\mathbf{B_0}$ and $x||\mathbf{B_1}$, write down the (time-dependent) Hamiltonian for the spin-1/2 system.

(b) Next, we'll write the state of the spin as a superposition of its $\hat{S}_z$ eigenstates: $|\chi\rangle = a(t)|0\rangle + b(t)|1\rangle$ where $\hat{S}_z|0\rangle = \hbar/2|0\rangle$ and $\hat{S}_z|1\rangle = -\hbar/2|1\rangle$. Using your Hamiltonian from (a) and substituting into the Schrodinger equation, derive two differential equations for $a(t)$ and $b(t)$. Show that they can be written in the following form:

$$\dot{a}(t) = i\frac{\omega}{2}\, a(t) + i\Omega \cos \nu t\; b(t) \tag{2}$$

$$\dot{b}(t) = -i\frac{\omega}{2}\, b(t) + i\Omega \cos \nu t\; a(t). \tag{3}$$

Find expressions for ω and Ω in terms of the magnetic fields and gyromagnetic ratios. Note that Ω is commonly called the "Rabi frequency."

(c) The above equations are difficult to solve as they are written (though you can certainly do it numerically!), and in many cases it is useful and appropriate to make approximations. An extremely important/useful/common technique is the "Rotating Wave Approximation" or RWA, which is valid for the common situation $\Omega << \omega, \omega \sim \nu$.[2] Substitute the following expression into the equations above:

$$a(t) = \alpha(t) e^{i\nu t/2} \qquad b(t) = \beta(t) e^{-i\nu t/2} \tag{4}$$

[1] I am designating a vector as (x-component, y-component, z-component).

[2] Here's the rough picture: for $\omega = \nu$ and $\Omega \to 0$, the solutions to this equation would only involve phase accumulation, with $b(t) = \beta e^{-i\omega t/2} \sim \beta e^{-i\nu t/2}$ and similar for $a(t)$. We understand that fast evolution, and we want to find out the effects of the oscillating field (which will be slower, since $\Omega << \omega$). So we want to write $b(t)$ as (slow variations in β)*(fast phase accumulation), and similar for $a(t)$.

and derive two differential equations for $\alpha(t)$ and $\beta(t)$. This gives you equations of motion in the so-called "rotating frame." Expand the cosine into exponentials, and show that

$$\begin{aligned}\dot{\alpha}(t) &= i\frac{\delta}{2}\,\alpha(t) + i\frac{\Omega}{2}\left(1+e^{-2i\nu t}\right)\,\beta(t)\\ \dot{\beta}(t) &= -i\frac{\delta}{2}\,\beta(t) + i\frac{\Omega}{2}\left(1+e^{2i\nu t}\right)\,\alpha(t).\end{aligned} \tag{5}$$

What is δ mathematically? Physically?

Now comes the magic: for $\Omega << \nu$, $\alpha(t)$ and $\beta(t)$ should vary slowly compared to the frequency ν. These differential equations each have a term $e^{\pm 2i\nu t}$ which oscillates extremely quickly in time and will average to zero over the slow timescales on which α and β evolve. So we can (approximately!) eliminate them to obtain

$$\begin{aligned}\dot{\alpha}(t) &= i\frac{\delta}{2}\,\alpha(t) + i\frac{\Omega}{2}\,\beta(t)\\ \dot{\beta}(t) &= -i\frac{\delta}{2}\,\beta(t) + i\frac{\Omega}{2}\,\alpha(t).\end{aligned} \tag{6}$$

(d) Solve E q. 6 for the initial condition that the spin starts out spin "up" (in state $|0\rangle$) at time $t = 0$, and find the probability to measure the spin to be "down" (in state $|1\rangle$) at time $t = \tau$. This is the probability of "flipping" the spin with a pulse of microwaves. Plot your result (i) as a function of time for $\delta = 0$ (ii) as a function of time for $\Omega = \delta$ (iii) as a function of δ for $\Omega\tau = \pi$. You should see "Rabi oscillations" in (i) and (ii) - what is their frequency?

(e) Just for this part, suppose that Ω depends on time in Eq. 6. Assume that the microwaves are "on resonance" ($\delta = 0$), and that the spin starts out in $|0\rangle$. Show that the spin state at time $t = \tau$ depends only on the area under the microwave pulse, $A = \int_0^\tau \Omega(t)dt$.

(f) Investigate the validity of the RWA. Using your favorite software, numerically integrate Eq. 5 and compare to your analytic results from Eq. 6. When is the RWA valid? When does it break down?

(g) You can get a lot of intuition by rewriting the Hamiltonian in the RWA. Eq. 6 has the following form:

$$i\hbar\frac{d}{dt}\begin{pmatrix}\alpha(t)\\ \beta(t)\end{pmatrix} = \hat{\mathbf{H}}_{\mathrm{RWA}}\cdot\begin{pmatrix}\alpha(t)\\ \beta(t)\end{pmatrix}. \tag{7}$$

What is $\hat{\mathbf{H}}_{\mathrm{RWA}}$? What does this remind you of? (See Problem 1(c)...) How could you use your results from Problem 1 to visualize the dynamics of a driven spin on the Bloch sphere? There is a simulator available at http://www.drcmr.dk/BlochSimulator/ which will allow you to switch between the laboratory and rotating frame to visualize the spin dynamics on the Bloch sphere. Note that they have already made the RWA by using a rotating microwave magnetic field instead of a linear, sinusoidally varying one. (Microwaves are "RF" in this simulation). See if you can figure out how the simulator works!

(h) What are the eigenstates of $\hat{\mathbf{H}}_{\mathrm{RWA}}$? This is the so-called "dressed-state basis" for a driven two-level system. What is the approximate form of these eigenstates for $\Omega << \delta$? For $\Omega >> \delta$? Plot the energy of these eigenstates for fixed Ω as a function of δ. What is the energy splitting at $\delta = 0$?

(i) How does this entire analysis change if $\mathbf{B_1}||y$? Determine $\hat{\mathbf{H}}_{\mathrm{RWA}}$ for $\mathbf{B_A C} = B_1(\cos\theta, \sin\theta, 0)\cos\nu t$. You should find it behaves exactly the same as if $\mathbf{B_A C} = B_1(1,0,0)\cos(\nu t - \theta)$! There's a very good reason for this: the RWA is equivalent to breaking up a (linear) AC field into a superposition of clockwise and counter-clockwise rotating fields...and then only keeping the component that is rotating in the same direction as the spin. Look at the simulation, draw some pictures, and try to convince yourself this makes sense. Also, check my signs...

(j) What happens if $\mathbf{B_1}||z$?

Note that this spin-1/2 analysis can be directly applied to the NV center provided that: (i) We spectrally isolate one spin transition from the others – for example, we can isolate the transition $m_s = 0$ to $m_s = 1$ (at frequency ω_+) from $m_s = 0$ to $m_s = -1$ (at frequency ω_-) by applying a static magnetic field B_z along the z axis. You saw this behavior in 1(d). (ii) The Rabi frequency Ω and the detuning δ must be small compared to difference between these transition frequencies, $\Omega, \delta << \omega_+ - \omega_-$. Thus we are tuning the microwaves close to one of the transitions, e.g. $\nu \approx \omega_+$, and ignoring their effect on the other transition. To map our spin-1/2 analysis onto the NV center, we need to associate $m_s = 0 \rightarrow |0\rangle$, $m_s = 1 \rightarrow |1\rangle$, and $\Delta - \gamma B_z \rightarrow -\gamma B_0$. However, there may be situations when we need to consider the full spin-1 system. Then another treatment is required using the spin-1 pauli matrices and a similar RWA approximation.

3. Pulsed excitation and the Ramsey sequence

(a) Using your results from problem 2, determine a propagator $\hat{U}$ for the driven spin-1/2 system in the RWA for $\mathbf{B_1} = B_1(\cos\theta, \sin\theta, 0)$. I highly recommend that you use Mathematica or a similar software to do this - the algebra can get intense. $\hat{U}$ should satisfy:

$$\begin{pmatrix} \alpha(t) \\ \beta(t) \end{pmatrix} = \hat{\mathbf{U}}(t) \cdot \begin{pmatrix} \alpha(0) \\ \beta(0) \end{pmatrix}. \tag{8}$$

In this case, you can solve Eq. 6 for $\alpha(0) = \alpha_0, \beta(0) = \beta_0$ and extract the propagator, but there are other more general approaches. Formally, from the Schrodinger equation $\hat{U} = e^{-i\hat{H}t/\hbar}$. To exponentiate an operator, one can use the Taylor expansion for an exponential; alternately, one can diagonalize the Hamiltonian $\hat{H}|\phi_n\rangle = \lambda_n|\phi_n\rangle$ and write $\hat{H} = \sum_n \lambda_n|\phi_n\rangle\langle\phi_n|$ so that $\hat{U} = \sum_n e^{-i\lambda_n t/\hbar}|\phi_n\rangle\langle\phi_n|$. *Answer:*

$$\hat{U} = \begin{pmatrix} \cos(\Omega_R t/2) + i\frac{\delta}{\Omega_R}\sin(\Omega_R t/2) & i\frac{\Omega e^{-i\theta}}{\Omega_R}\sin(\Omega_R t/2) \\ i\frac{\Omega e^{i\theta}}{\Omega_R}\sin(\Omega_R t/2) & \cos(\Omega_R t/2) - i\frac{\delta}{\Omega_R}\sin(\Omega_R t/2) \end{pmatrix} \tag{9}$$

where $\Omega_R = \sqrt{\Omega^2 + \delta^2}$. Note that $\hat{U}$ is not Hermitian but it is unitary, $\hat{U}^\dagger\hat{U} = \hat{1}$.

Unfortunately, the propagator derived here is only valid for constant Ω, but it can nevertheless be used to understand spin evolution when driven by microwave pulses. For example, $|\chi\rangle = \hat{U}(\Omega e^{i\theta}, \delta, \tau)|\chi(0)\rangle$ describes the spin state after a single pulse of microwaves with Rabi frequency Ω, phase θ, detuning δ, and duration τ. To describe two pulses of duration τ separated by a pause of duration t we would write

$$|\chi\rangle = \hat{U}(\Omega e^{i\theta}, \delta, \tau) \cdot \hat{U}(0, \delta, t) \cdot \hat{U}(\Omega e^{i\theta}, \delta, \tau)|\chi(0)\rangle. \tag{10}$$

Adding on more pulses amounts to matrix multiplication. Note that the pulse on the right is applied first.

(b) Find $\hat{U}$ for (i) $\delta = 0, \theta = 0, \Omega t = \pi/2$ and (ii) $\delta = 0, \theta = 0, \Omega t = \pi$. These describe a resonant "$\frac{\pi_x}{2}$ pulse" and a "π_x pulse" respectively. What do they do to the spin on the Bloch sphere (in the rotating frame)?

(c) Find $\hat{U}$ for (i) $\delta = 0, \theta = \pi/2, \Omega t = \pi/2$ and (ii) $\delta = 0, \theta = \pi/2, \Omega t = \pi$. These describe a resonant "$\frac{\pi_y}{2}$ pulse" and a "π_y pulse" respectively. What do they do to the spin on the Bloch sphere (in the rotating frame)?

(d) In the limit that $\Omega >> \delta$ (so-called "hard pulses"), the matrices you found in (b) and (c) very closely approximate the actual behavior even with finite detuning. Use this approximation to model the behavior of a spin during a so-called "Ramsey sequence" of $\frac{\pi_x}{2}$ − wait − $\frac{\pi_x}{2}$ with a small but finite detuning δ of the microwave pulses. In particular, find and plot the probability to measure the spin to be in state $|1\rangle$ if it starts off in state $|0\rangle$. How would this look different if you used $\frac{\pi_x}{2}$ − wait − $\frac{\pi_y}{2}$?

(e) How good is the hard-pulse approximation? Examine the Ramsey sequence using the full expression for the detuned pulses (as in Eq. 10) and compare to the approximate expression you found in (d).

Suggested references:

- Griffths, Introduction to Quantum Mechanics (or whatever quantum textbook you used).
- Metcalf and van der Straaten, Laser Cooling and Trapping (first 2 chapters)
- Slichter, Principles of Magnetic Resonance, pp1-4, 11-13, 17-19, 20-22, 31-33.
- Scully and Zubairy, Quantum Optics, pp151-155.
- R. Hanson, O. Gywat, and D. D. Awschalom. Room-temperature manipulation and decoherence of a single spin in diamond. Physical Review B 74: 161203 (2006)–(especially the first page and a half)
- Physics 285b Chapter 2, `http://lukin.physics.harvard.edu/wp-uploads/Papers/285b_notes_2005-1.Lily.pdf`